

## L2 – 3.3 Quotient of Linear Functions

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### Part 1: Key Features of the Quotient of Linear Functions

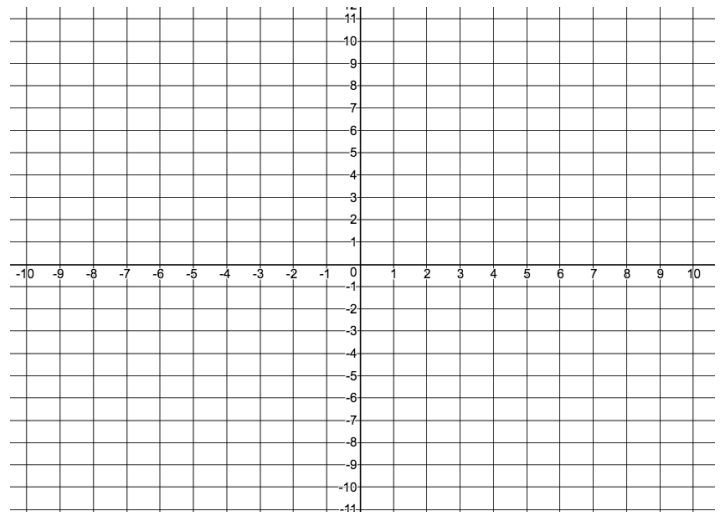
Features of  $f(x) = \frac{ax+b}{cx+d}$

- If an  $x$  value is a zero of the denominator ONLY, this results in a vertical asymptote
  - Equation of vertical asymptote is
- If an  $x$  value is a zero of the numerator AND denominator, this results in a \_\_\_\_\_ in the graph NOT a vertical asymptote
- There is a horizontal asymptote at the ratio of the leading coefficients
  - Equation of horizontal asymptote is
- Forms a \_\_\_\_\_: the two branches of the graph of the function are equidistant from the point of intersection of the vertical and horizontal asymptotes
  - Once you know the shape of one branch, you can translate the points to graph the other branch
- You can find the  $x$ -intercept by setting  $y = 0$  and solving for  $x$ 
  - This results in
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### Part 2: Graphing a Quotient of Linear Functions

**Example 1:** Graph each of the following functions

a)  $f(x) = \frac{x-3}{x+2}$



**b)**  $g(x) = \frac{2x-3}{x-1}$

